

12	(a)	(i)	e.g. satellite <u>communication</u> , mobile phones, line of sight communication, wifi	B1	[1]
		(ii)	e.g. connection of TV to aerial, loudspeaker, microphone (if clearly identified)	B1	[1]
		(iii)	e.g. a.f. amplifier to loudspeaker, landline for phone	B1	[1]
	(b)	(i)	attenuation/dB = $10 \lg (P_2/P_1)$ $-190 = 10 \lg (P_2/3.1)$ $P_2 = 3.1 \times 10^{-19} \text{ kW}$	C1	
				A1	[2]
		(ii)	signal is amplified frequency is changed to prevent swamping of up-link signal by down-link (signal)	M1	
				M1	
				A1	[3]

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